Learning manifolds with variational autoencoders

Weekend 4, Friday morning

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CAS Building ML/AI Applications, ETH Zurich

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Three numbers per picture



Three numbers per picture



$\longrightarrow (x, y, m)$

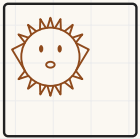
x, y: where the fish is

m: how puffed up it is



$m = 0$ $m = 0.5$ $m = 1$

Three numbers per picture



$(x, y, m) = (0.35,\ 0.60,\ 1.00)$



$(x, y, m) = (0.68,\ 0.62,\ 0.50)$



$(x, y, m) = (0.45,\ 0.25,\ 0.00)$

x, y: where the fish is

m: how puffed up it is



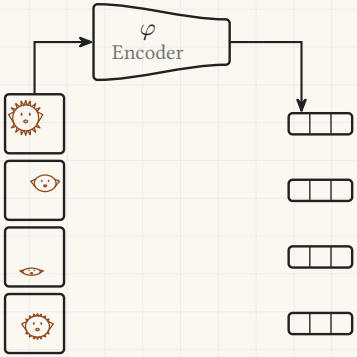
$m = 0$ $m = 0.5$ $m = 1$

The yogurt-cup game



Pictures

The yogurt-cup game

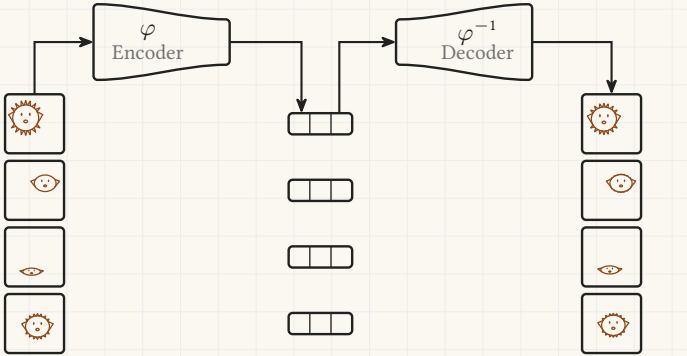


Pictures

Representations

Three numbers per picture

The yogurt-cup game



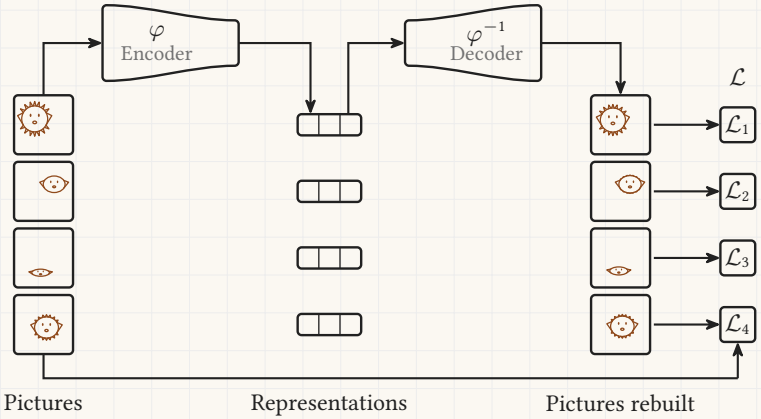
Pictures

Representations

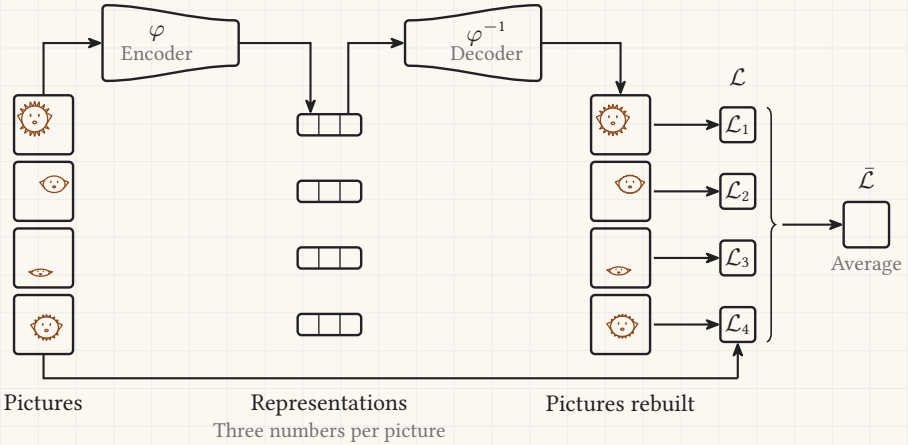
Three numbers per picture

Pictures rebuilt

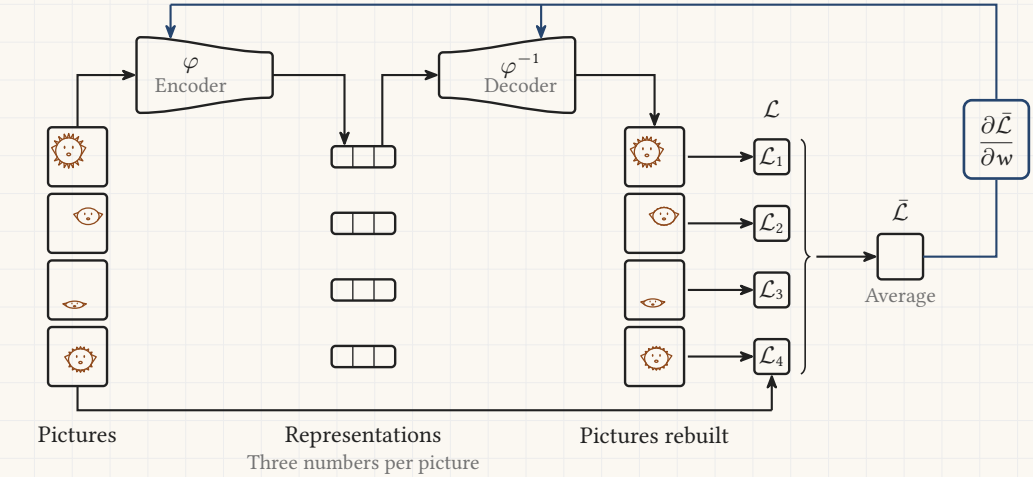
The yogurt-cup game



The yogurt-cup game



The yogurt-cup game

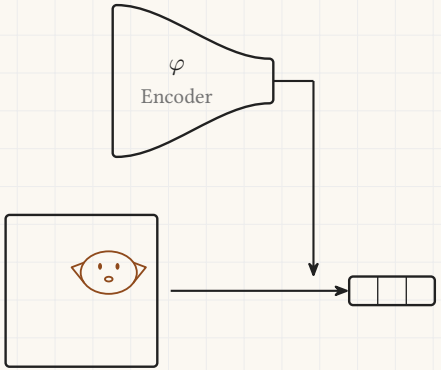


Changing the representation

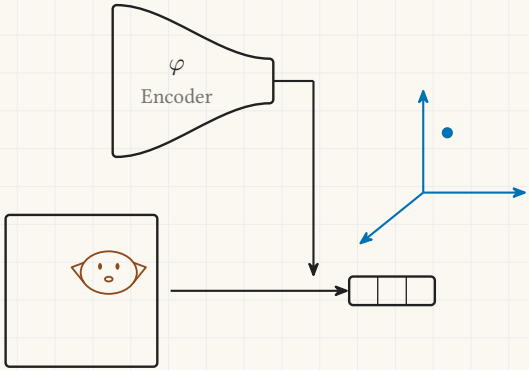


Picture

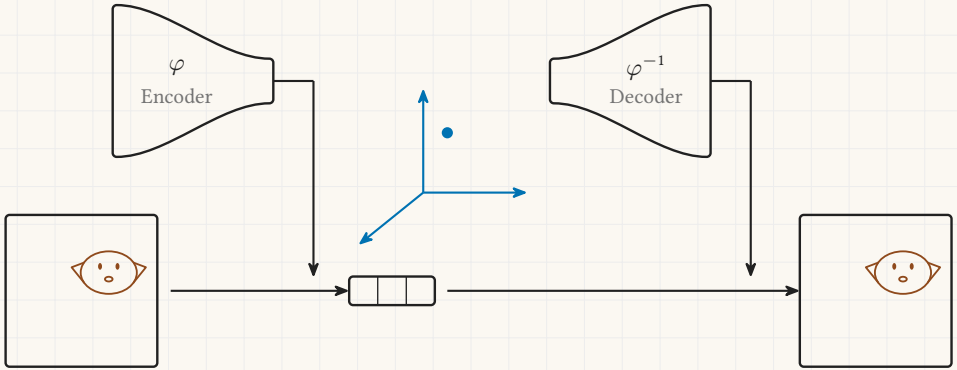
Changing the representation



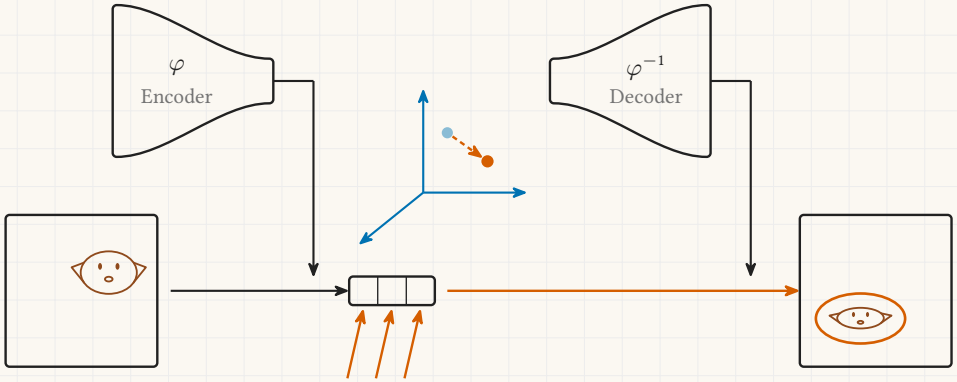
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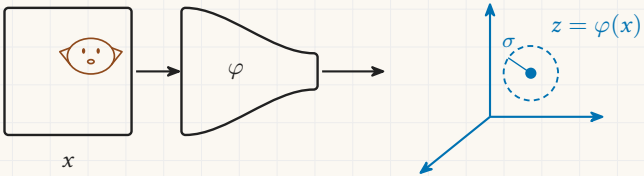
The pufferfish notebook



1. 5000 made-up pufferfish pictures, each drawn from its three numbers (x, y, m).
2. A plain autoencoder squeezes every picture into three numbers of its own, and back.
3. Your part: fill in the three gaps marked ◎, then train it, about a minute and a half on Colab.
4. The last cell is the previous slide, live: pick a fish, change its three numbers, see the new fish.

`https://colab.research.google.com/github/eth-bmai-hs26/w4-cx-public/blob/main/pufferfish-autoencoder/pufferfish_autoencoder.ipynb`

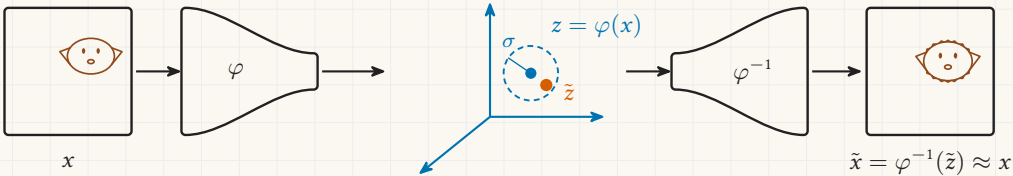
Variational autoencoders



1. The encoder gives each picture a centre $z = \varphi(x)$ and a radius σ.

Kingma and Welling, Auto-Encoding Variational Bayes, ICLR 2014

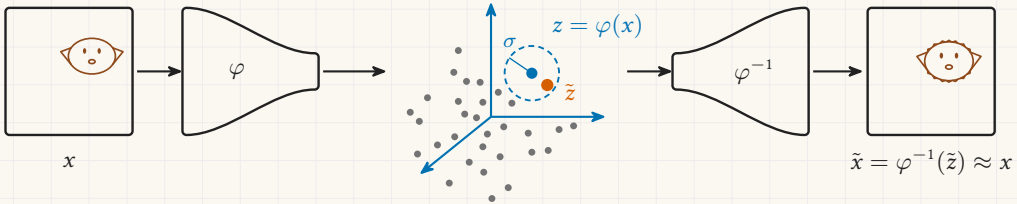
Variational autoencoders



1. The encoder gives each picture a centre $z = \varphi(x)$ and a radius σ.
2. The decoder gets a random $\tilde{z}$ within about σ of z, and must still rebuild x.

Kingma and Welling, Auto-Encoding Variational Bayes, ICLR 2014

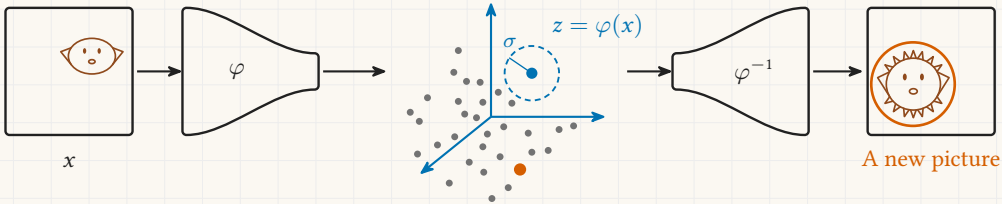
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2. The decoder gets a random $\tilde{z}$ within about σ of z, and must still rebuild x.
3. A second term in the loss pulls the codes of all pictures into one round cloud.
4. To make a new picture, pick any point in the cloud and decode it.

Kingma and Welling, Auto-Encoding Variational Bayes, ICLR 2014